Quiz2

Name: **Admission No:**

Examination: **Monsoon 2025-26** Marks:**10**

Subject: **Computer Programming (NCSV101)** Time: **15 Minutes**

Instructions:

Answer all the questions. All the questions are single answer correct only. **Assume that all the required header files are included. Assume the compiler does not require return 0 at the end of main functions below.** **10 * 1 = 10 marks**

Q1. What will be the output of the following code?

```
int main()
{
  int x[] = {6,2,10,11,5,-1};
  int *p = x;
  int a = *p++;
  int b = *p;
  printf("%d %d",a,b);
}
```

A. **6 2**
B. 2 2
C. 7 6
D. 6 6

Q2. What will be the output of the following code?

```
int fun(int n){
   if(n%3 == 0) return 2+fun(n+1);
   else if(n%3 == 1) return 1+fun(n+1);
   else return n;
}
int main(){
   printf("%d", fun(6));
}
```

A. 3
B. 6
C. 9
D. **11**

Q3. In C, if a function definition appears after the main() function, what must be included before main() to avoid compilation errors?

A. A macro definition
B. **A function prototype**
C. A header file inclusion
D. A preprocessor directive

Q4. What will be the output of the following code snippet?

```
int arr[5] = {10, 20, 30, 40, 50};
int *p1 = &arr[1];
int *p2 = &arr[4];
printf("%d", p2 - p1);
```

A. 2
B. **3**
C. 4
D. Undefined behaviour

Q5. What is the correct output of the following code?

```
int main(){
int a = 0;
int *p = &a;
if (p)
    printf("Yes");
else
    printf("No");
}
```

A. **Yes**
B. No
C. Undefined
D. Depend on the system

Q6. What is the correct output of the following code?

```
int fun(int x) {
    x += 10;
    return x;
}
int main() {
    int a = 5;
    fun(a);
    printf("%d", a);
}
```

A. 15
B. 10
C. **5**
D. None of these

Q7. What is the correct output of the following code?

```
int main(){
      struct S { int arr[3]; };
      struct S s = {{1, 2, 3}};
      printf("%d", s.arr[1]);
      return 0;
}
```

A. Compilation error
B. Undefined
C. 1
D. 2

Q8. Consider the following C code fragment:

```
float numbers[5] = {2.5, 3.5, 4.5,
5.5, 6.5};
float* arr = numbers;
x = (*arr)++;
y = *(arr++);
```

Which of the following can be deduced?

A. While x is a float value, y is an address
B. While x is an address, y is a float value
C. Both x and y are float values
D. Both x and y are addresses

Q9. Predict the output of the following C program:

```
#include<stdio.h>
int next(int after)
{
    int i = after+1, j;
    for(j = 2;;i++)
    {
        while(i%j++ != 0);
        if(j > i)
            return i;
        else
            continue;
    }
}
int main()
{
    printf("%d,%d\n", next(8), next(9));
    return 0;
}
```

A. 8,9
B. 9,9
C. 11,11
D. 7,7

Q10. Predict the output of the following C program:

```
#include<stdio.h>
void func(int i, char c)
{
    static int j;
    j = c == 'Y' || c == 'y' ? 1 : j;
    if(i == 1)
        printf("%d\n", j);
    else
    {
        j *= i;
        func(i-1,'n');
    }
}
int main()
{
    func(5, 'Y');
    return 0;
}
```

A. 2
B. 120
C. Compilation Error
D. No Output (infinite execution)